

Barcode Lists — Debug Guide

Companion to BL-SOP-001 · For IT / Support / Developer use

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Version	1.1
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Applies to	Extension v7.11 (Manifest V3, no build step)
Ordering website	Lexmark — automation must start at the ALL BATCHES page

What changed in v1.1: ordering website named **Lexmark**; the required **ALL BATCHES** start page and the "**Content not found**" fix are documented as first-line user remediations; new `tcoResume` storage key (mid-run page refresh with automatic resume, v7.10); the TCO Excel fixed-column-H dependency and stale-config migration (v7.11) added to Known Issues; two new failure recipes (8.5, 8.6).

1. Architecture in 60 Seconds

<code>popup.js</code>	App core: in-memory <code>state</code> (categories, comments, active, insertedBarcodes, unopenedCategories, importantCategories, trash), renders side panel, all UI events. Persists to <code>chrome.storage.local</code> then pushes to Supabase per-item when online. Also builds automation batch payloads and validates XPath's before launch.
<code>supabase.js</code>	Raw <code>fetch()</code> to PostgREST. Auth = store number + DJB2 base36 password hash. Session in <code>chrome.storage.local.supabaseSession</code> .
<code>background.js</code>	MV3 service worker: opens panel on icon click, 1-min polling alarm, NEW badge + notification on remote changes.
<code>content.js</code>	Runs on all URLs: executes auto-add / TCO automation on the Lexmark ordering site, search-box insertion, green overlay. Persists a <code>tcoResume</code> checkpoint before the mid-run <code>location.reload()</code> and resumes from it after the reload (120 s TTL).
<code>xlsx.full.min.js</code>	Vendored SheetJS for Excel import.

No build step. Source IS the extension.

2. DevTools Access

Surface	How to open
Side panel UI (popup.js)	Right-click inside the panel → Inspect. Or <code>chrome://extensions</code> → Barcode Lists → "Inspect views: side panel".
Service worker (background.js + supabase.js network)	<code>chrome://extensions</code> → Developer mode ON → Barcode Lists card → click service worker link. Note: MV3 workers sleep; opening the panel or the link wakes them.
Content script (content.js)	DevTools on the Lexmark tab itself. Logs prefixed <code>[BarcodeLists]</code> .

3. Storage Inspection

Panel DevTools → Application → Storage → Extension Storage → Local. Or console: `chrome.storage.local.get(null, console.table)`

Key	Contents
<code>barcodeData</code>	Main state: <code>{ categoryOrder, categories: {name: [{barcode, comment},...]}, comments: {barcode: text}, active, importantCategories, insertedBarcodes, unopenedCategories, trash... }</code>
<code>supabaseSession</code>	<code>{ storeNumber, storeId, passwordHash }</code> — survives browser restart until logout (local, not session, storage).
<code>autoAddConfig</code>	XPath + delay/timeout settings (defaults in <code>DEFAULT_AUTO_ADD_CONFIG</code> , <code>popup.js:6</code>). v7.11 migrates stale saved TCO XPaths to the corrected values on load.
<code>dontShowRunToast</code>	Boolean — "Don't show this again" state for the run pop-up.
<code>trashData</code>	Deleted lists awaiting restore.
<code>lastBarcodeCount</code> / <code>lastTimestamp</code>	Background poll watermarks for NEW badge detection.
<code>tcoResume</code> NEW v1.1	TCO automation checkpoint written just before <code>location.reload()</code> : <code>{ batches, config, batchIndex, phase:"afterOpen", curAdded, curSkipped, totalAdded, totalSkipped, totalEdited, totalEditSkipped, ts }</code> . Consumed on the reloaded page to resume mid-batch; TTL 120 000 ms (<code>TCO_RESUME_TTL_MS</code> , <code>content.js:111</code>) — stale checkpoints are discarded and the run must be restarted from ALL BATCHES.

Nuclear reset (destroys local unsynced data): remove the extension via `chrome://extensions`, or `chrome.storage.local.clear()` from the panel console. Cloud data re-pulls at next login.

4. Console Message Reference

String	Source	Meaning
Sync failed, using local data: ...	popup.js:130	Login-time pull failed → offline fallback
Auto-sync failed: ...	popup.js:169	Pull on NEW_DATA message failed
chrome.storage not available	popup.js:256	Storage API missing (context invalidated after update/reload)
Excel processing error: ...	popup.js:1259	SheetJS parse failure
Login error: <status> ...	supabase.js:34	REST error on stores lookup
Poll error: ...	background.js:74	Background poll fetch failed (offline)
Notification error: ...	background.js:107	notifications.create rejected
[BarcodeLists] typed batch name / clicked create batch button / selected all signs in batch / applied template / saved TCO price for: ...	content.js	Automation step trace — the last line before a failure identifies the broken step
[BarcodeLists] resuming TCO automation after refresh NEW v1.1	content.js:623	Checkpoint consumed; run continues from the saved batch/phase after the mid-run <code>location.reload()</code> . Absent after a reload ⇒ checkpoint expired (TTL 120 s) or was never written.

Silent-failure design: remote write errors in CRUD calls are `.catch(console.error)` only — the UI stays fully functional on local state. If a user reports "changes vanished on another device", check the panel/service-worker console for these errors first.

5. Runtime Message Map

Channel	Messages
background → popup	NEW_DATA (remote change detected)
popup → background	CLEAR_BADGE
popup → content	AUTO_ADD_BATCH_START , AUTO_TCO_BATCH_START , AUTO_ADD_STOP , SEARCH_BARCODE , SHOW_OVERLAY
content → popup	AUTO_ADD_PROGRESS , AUTO_ADD_BATCH_STARTING/DONE , AUTO_ADD_DONE , AUTO_TCO_EDIT_PROGRESS , AUTO_TCO_DONE , AUTO_ADD_STOPPED , AUTO_STATUS , AUTO_ADD_ERROR , SEARCH_BARCODE_DONE/ERROR

TCO resume flow NEW v1.1

1. AUTO_TCO_BATCH_START → per batch: add loop → open batch breadcrumb → write tcoResume → location.reload() .
2. Content script re-injects on the fresh page, reads tcoResume (valid < 120 s), logs "resuming TCO automation after refresh", skips the completed add loop, proceeds to select-all → template → per-sign price edits.
3. On finish/expiry/stop the checkpoint is removed (clearTcoResume).

There is no popup involvement in the resume — it is entirely content-script + storage. A user-reported stall "after Refreshing batch..." almost always means the checkpoint expired (> 2 min on the reload screen) or the reload landed on a page other than the opened batch (see 8.5).

6. Backend (Supabase)

Project `https://jahcgdgm dasrhzcuy nkw. supabase. co` · anon JWT hardcoded (supabase.js:2, exp 2036) · RLS policies wide-open (`USING (true)`) — isolation is per-store by convention, not enforced.

Table	Purpose / notes
<code>stores</code>	<code>store_number</code> unique, <code>password_hash</code> = DJB2 base36 (init 0, <code>((h<<5)-h)+c</code> per char), <code>updated_at</code> touched on writes. Auto-created on first login (POST).
<code>barcodes</code>	<code>store_id</code> , <code>category_name</code> , <code>barcode_value</code> , <code>created_at</code> . Core data.
<code>barcode_comments</code>	≤250 chars, <code>UNIQUE(store_id, barcode_value)</code> , upsert merge-duplicates. DDL: <code>supabase _comments. sql</code> .
<code>important_categories</code>	Write-only from phone app; extension reads + reconciles (never writes new rows except name-rename PATCH).
<code>category_order</code>	Position per list; delete-all + reinsert on rename/order change.
<code>copied_barcodes</code>	Purple "searched" state (<code>copied: true</code>).
<code>opened_categories</code>	Green-glow clearing (upsert on open). No migration in repo.

Sync strategy: pull at login (`syncFromRemote`) + on `NEW_DATA` ; writes are per-item CRUD (`addBarcodeRemote` , `removeBarcodeRemote` , `clearCategoryRemote` , `renameCategoryRemote`). Note: `syncToRemote` (delete-all-reinsert, supabase.js:455) exists but is dead code — not called from popup.js. Online probe = live GET on `stores?select=id &limit=1` (`isOnline()` supabase.js:88); offline gate = `isOnlineMode` .

7. Known Issues & Doc Mismatches

Item	Detail
Web Store version lag	Store serves 7.7; internal is 7.11. 7.7 lacks yellow TCO broom (7.8), inline rename (7.9), mid-run refresh + resume (7.10), corrected TCO XPathS + config migration (7.11). First-line fix for "automation is broken": update (SOP §4 — always check for updates).
Store listing overstates features	"Photo/OCR import" and "AI cleanup" described on listing are not in current code.
README/AGENTS mismatches	Session storage claimed <code>chrome.storage.session</code> , actual local; copy-to-clipboard replaced by search icon; <code>syncToRemote</code> described as strategy but is dead code; "syncing" yellow dot class never set (only online/offline); <code>showLoadingOverlay</code> defined but unused.
Security posture	Hardcoded anon key + open RLS; DJB2 is trivially reversible; no rate limiting on store creation. Acceptable for internal use; do not expose publicly.
Automation fragility	XPaths target one Lexmark layout; site redesigns break runs with "... not found" toasts. Fix = update <code>autoAddConfig</code> (Settings) or defaults in <code>popup.js:6</code> . "Content not found" (Lexmark page error) is different: wrong start page — user must be at ALL BATCHES and refresh; not an XPath bug.
TCO Excel fixed column H NEW v1.1	<code>hasNoChangeComment()</code> (<code>popup.js:1169</code>) reads column index 7 (column H) by position , regardless of headers: any row whose column-H text contains the word <code>no</code> or <code>change</code> is skipped on import. UPC header must match <code>/\bupc\b/i</code> within the first 10 rows (<code>popup.js:1148</code>); price header must contain both <code>tco</code> and <code>price</code> (<code>popup.js:1178</code>). Wrong header names, shifted columns, or bad cell values are the dominant failure mode of the TCO flow — human error, not code. Debug: open the source file, verify per SOP §8.1.
Stale saved configs NEW v1.1	Clients upgrading to 7.11 carry pre-7.11 TCO XPathS in <code>autoAddConfig</code> that point at dead elements. v7.11 migrates them automatically on load — if a client still fails on a fresh install, check whether a manually saved config overrides defaults.

8. Failure Recipes

8.1 "Sync broken" report

1. Panel console: look for `Sync failed...` / `Auto-sync failed...` → read status (CORS? 401? network?).
2. Service worker console → Network tab → repeat action → inspect REST call + response body.
3. Verify session: `chrome.storage.local.get('supabaseSession')`.
4. Verify remote row: `GET /rest/v1/barcodes?store_id=eq.<id>&order=created_at` with apikey header.

8.2 Automation fails at a step

1. Lexmark-tab DevTools console → last `[BarcodeLists]` line names the step.
2. Confirm the run started from **ALL BATCHES** (most "not found" reports) — if not, user fix, no code change.
3. Inspect the corresponding element; write new XPath/CSS selector.
4. Update Settings (user-level) — or `DEFAULT_AUTO_ADD_CONFIG` for a shipped fix.
5. Re-test with `test-automation.html` locally before real runs.

8.3 "Extension context invalidated" after update

Old panel instance still open post-reload. Close/reopen the side panel. If persistent: `chrome://extensions` → reload card. Local storage survives; session survives.

8.4 Password reset (no self-service)

1. Compute hash: `node -e "let h=0;for(const c of 'NEWPASS'){h=((h<<5)-h)+c.charCodeAt(0);h&=h}console.log(h.toString(36))"`
2. `PATCH /rest/v1/stores?store_number=eq.FMCXX` body `{"password_hash":"<hash>"}` with service/admin key.
3. User logs in with new password; existing sessions keep working until logout.

8.5 TCO run stalls at "Refreshing batch..." NEW v1.1

Expected: the content script wrote `tcoResume` and called `location.reload()` (v7.10 behaviour).

1. On the fresh page, console must show `[BarcodeLists]` resuming TCO automation after refresh (content.js:623). Absent ⇒ checkpoint expired (reload idle > 120 s) or never written (check for `could not persist TCO resume state` — storage failure).
2. Inspect `chrome.storage.local.get('tcoResume')`: `phase:"afterOpen"`, correct `batchIndex`, counters non-zero if the add loop ran.
3. If resume fired but select-all fails, the reload landed on a non-batch page — verify the user's `tcoOpenBatchXPath` breadcrumb and that Lexmark was left untouched during the refresh.
4. Recovery: delete `tcoResume`, put Lexmark at **ALL BATCHES**, refresh, re-run. TTL guarantees no duplicate price edits from a stale checkpoint.

8.6 TCO prices missing or wrong on signs NEW v1.1

1. **Source file first.** Verify against SOP §8.1: `UPC` header (`\bupc\b` , first 10 rows), `TC0 Price` header (contains both words), values numeric, no/change notes in **column H**. Human error here is the dominant cause — check it before blaming code.
2. Confirm comments landed: list rows show `TC0 Price: x.xx ; DB: barcode_comments` rows match `/^tco\s*price:\s*/i` (popup.js:1645). Missing ⇒ import skipped the rows (no/change note, <5 digits after cleaning, or bad header).
3. Signs skipped in edit pass = no TCO comment on that barcode (by design; reported in the "N skipped" summary).
4. Run trace: last `[BarcodeLists]` saved `TC0 price for:` line names the last successfully edited barcode; the error message names the failing field (`Price field not found` ⇒ `tcoPriceFieldXPath` , default `#7054`).

